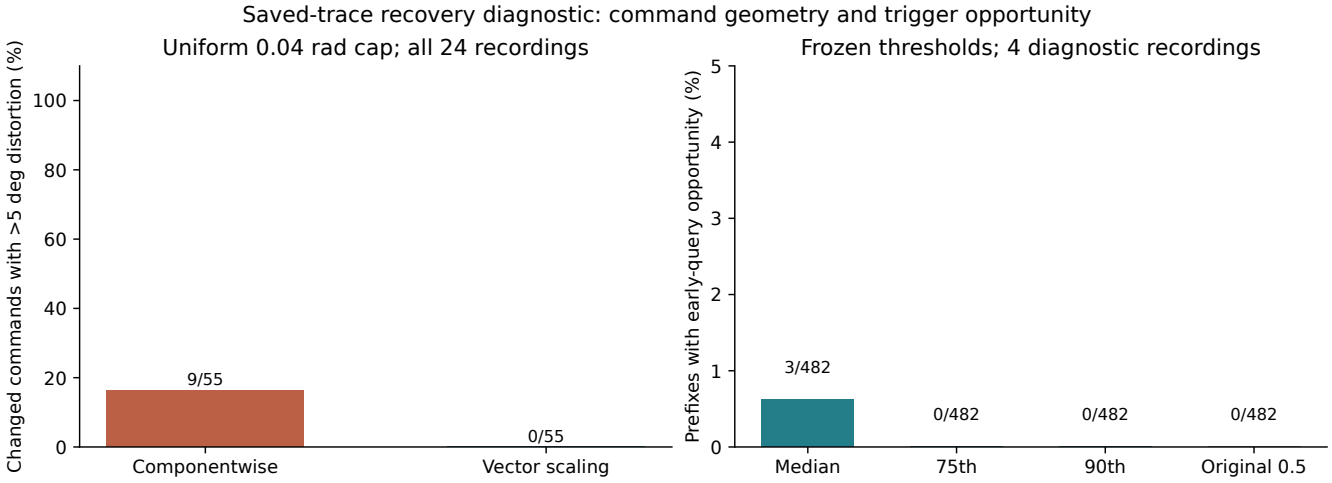
SAFETY DIAL / SUPERVISOR CHECK-IN / 19 OCTOBER 2026

A working repair, with the physical benefit still to test

Sachin Mohan, 2699183 Supervisor: Geraud Nangue Tasse

Research question. Can evolutionary search find a compact execution controller that improves safe completion around a frozen manipulation policy, under matched interaction budgets?

Completed today. Implemented and tested a uniform cap that scales the whole seven-joint target-error vector, preserving its direction. Ran a frozen saved-trace diagnostic on **18,657 commands from 24 historical episodes**. Prepared a paper with three algorithms, explicit controller equations and audited outcomes, plus labelled historical demonstration videos.



Concrete new result

At a .04 rad cap, componentwise clipping rotates joint-error direction by more than 5° on **9/55** modified commands. Vector scaling has **0/55**, with maximum numerical deviation 0.00019°. This verifies a command-geometry property; it does not measure physical safety.

What limits the timing mechanism

The median threshold .089753 admits **3/482** early-query opportunities, all on one of two diagnostic tasks. Calibration has only five positive eligible values. These are correlated recorded-prefix opportunities, not executed queries or a new safety result.

Physical evidence retained, including the adverse result

Comparison	Safe completion	Violating episodes	Interpretation
Part 1 identity	5/8	0/8	Original policy reference
Part 1 full evolved dial	4/8	2/8	Training lead did not transfer
Part 2 identity / cap-8 / adaptive	3/4 each	1/4 each	Adaptive rule never refreshed early
Part 2 cap-4	2/4	1/4	More queries did not improve this pilot

Next decision. Run the recovery plan's eight-episode mechanism screen: two preselected development tasks crossed with identity, componentwise cap/8, vector cap/8 and vector cap with active refresh. Freeze the cap, threshold and cases first. A later contribution claim needs matched random search and ordinary learning, independent optimizer seeds and reserved evaluation tasks.

Meeting materials. Open the paper draft for methods and full interpretation. The video package contains a selected full-dial success, an all-settings-safe case and an adverse case, retaining both cameras and native timing. Examples were chosen after outcomes; the pooled result remains above.

Scope: this machine has no installed ManiGuard simulation/policy stack; the user selected saved-trace work. Today's diagnostic adds zero simulator controls, physics steps, VLA queries, encoder calls or trained models. The native monitor labels violations and does not filter actions. No new physical improvement or optimizer superiority is claimed.